

Performance Testing

Date	10 July 2026
Team ID	SWTID-2026-8977
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing & API Response Testing
Target Module / API	FastAPI Backend APIs (Question Answering, Quiz Generation, Text Summarization, Learning Path Recommendation)
Test Environment	Local System (Windows 10, FastAPI, Google Gemini API)
Test Date	10 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
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1	Intelligent Question Answering API	10	60	AI returns accurate educational response within acceptable response time
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2	Quiz Generation API	10	60	Quiz generated successfully without errors
3	Text Summarization API	15	60	Summary generated correctly with minimal delay
4	Personalized Learning Path API	10	60	Learning roadmap generated successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Fast response for most requests
2	Response Time (Max)	< 5 seconds	3.2 seconds	Pass	Slight increase for complex AI prompts
3	Throughput (Req/sec)	> 15 Req/sec	18 Req/sec	Pass	Stable request handling
4	Error Rate	< 1%	0.3%	Pass	Minor timeout due to external API latency
5	CPU Utilization	< 80%	52%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	58%	Pass	Memory usage within acceptable limits

Step 4: Observations & Analysis

- ❑ The FastAPI backend handled concurrent requests efficiently with stable response times.
- ❑ Google Gemini API generated accurate educational responses for all tested modules.
- ❑ Quiz generation, concept explanation, and text summarization performed reliably during testing.
- ❑ CPU and memory utilization remained well below the defined threshold.
- ❑ Occasional response delays were observed for complex prompts due to network latency and external API processing time.
- ❑ Overall system performance was stable, responsive, and suitable for educational use.



Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output)

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